

Pavements, Energy Efficient Actors In Public Lighting

Sébastien Liandrat^{1,*}, Valérie Muzet², Vincent Bour³, Jérôme Dehon⁴, Jean-Pierre Christory⁵,
Brice Delaporte⁶, Joseph Abdo⁷, Florence Pero⁸, and Thibaut Le Doeuff⁹

¹ Corresponding author; Cerema ITS Research Team, Clermont Ferrand, France, sebastien.liandrat@cerema.fr

² Cerema ENDSUM Research Team, Strasbourg, France

³ AFE, Roissy, France

⁴ Schröder, Liège, Belgique

⁵ Consultant, Rambouillet, France

⁶ Routes de France & Office des asphaltes, Paris, France

⁷ Cimbéton, Paris, France

⁸ Specbea, Paris, France

⁹ CERIB, Épernon, France

ABSTRACT. The variety of road surfacing techniques meets most of the current challenges, particularly those related to physical and mechanical aspects such as skid resistance or durability. However, the light reflection capabilities of pavements are rarely considered in the design of lighting installations, as typical characteristics that are used are now being questioned.

The working group “Pavements and Lighting” is composed of many actors, both public and private, from very different background related to these topics. It is conducting a study that aim to develop tools and methods for managers, lighting designers and road builders to optimize lighting both in interurban and urban areas. The objective is to increase the knowledge on the photometric characteristics of pavements, the relationship between lighting and pavements and reduction of light pollution or urban heat phenomenon. One main deliverable of this study is a library with a complete characterization of the photometry of a large panel of current and innovative pavements available on the French market.

This library of urban and interurban pavements and the methodological approach will be available to all.

A series of 39 pairs of samples were tested with Cerema’s gonioreflectometer in the initial state and after 30 months of aging in outdoor conditions, exposed to rain and sun. This panel includes different surfacing materials: asphalt concrete (bituminous or synthetic binder), mastic asphalt, poured and precast cement concrete and natural stones.

The results show an enormous variability of photometric characteristics among the existing paving techniques as well as a significant evolution over time for some of them.

Lighting calculation were carried out to assess the impact of using actual photometric properties of the pavement in the lighting design. They demonstrated the importance of using the real pavement measurement rather than typical characteristics to avoid excessive lighting or poor uniformity, which can lead to potential road safety problems. Thus, a lighting installation optimized by considering a clear and diffusive pavement can lead to energy savings of up to 50% while maintaining optimal lighting quality, which highlights the importance of considering the pavement and light combination.

Keywords. Pavement Surface Properties, Road Photometry, Durability and Long-Term Performance of Pavements, Energy Efficiency

Introduction

Pavements and lighting context

A few decades ago, roads were only required to be robust, safe and comfortable for vehicle traffic. It was also always important that it remains economically acceptable, both during its construction and during maintenance phases. Over time, the number and variability of pavement families and techniques have also increased considerably to meet today's needs and uses. Thus, among the major families such as asphalt concrete, mastic asphalt, poured cement concrete, precast concrete and natural stone, there are still many variables. Aggregates, which are the essential component of many families and formulas, can vary in size, color and origin (natural, recycled, artificial) and then the particle size distribution can also differ. The binder can also vary, whether it is as often based on cement or bitumen, the additives can thus bring them different properties. New binders are also developed, sometimes synthetic or even of vegetable origin. Today, the road is still widely used and is asked to meet more and more societal needs and challenges. Indeed, the qualities described above such as comfort and safety have become prerequisites complemented by other objectives to be achieved.

Public lighting is one of these issues that refers both to traditional objectives in terms of safety and comfort of users but also to more recent societal concerns such as energy consumption and the issue of light pollution. The lighting installations are designed on the principle of "projected light flux" on a standard road surface. The important thing for the user is not only the projected luminous flux, but also the light reflected by the surface or the obstacle and thus the luminance. Perception is based on luminance contrast and this depends on the properties of the road surface. Road lighting installations are generally designed by calculating the performance in terms of luminance distribution as defined in (CIE 140, 2000) and the EN 13201 European standard (EN 13201-3:2015). Since the photometric characteristics of pavements are generally not measured, a standard r-table as defined in (CIE 144, 2001) is often used for lighting design. The "standard" r-tables are over 50 years old and several studies have shown that they are no longer representative of today pavements (Dumont et al., 2007, Jackett and Frith, 2010, Gidlund et al., 2019, Muzet, Greffier, et al., 2019).

Pavements and light work group

The *Pavement and Lighting* working group (P&L group or in French "Revêtements et Lumière") is composed of project managers and public authorities¹, professional associations and unions of lighting designers² and road builders³, public and private research organizations⁴, and expert consultants. After having organized and monitored demonstrators and operations on real sites to show the relevance of the issues and concepts of optimal lighting (Abdo et al., 2010, Christory et al., 2014), our objective is to develop a library of real and innovative urban pavements available on the market. This library is intended to facilitate the choice of decision makers and develop tools and methods for managers, lighting designers, and road builders to optimize lighting in interurban and urban areas.

The objective of this study was first to improve the knowledge of the photometric properties of a large number of urban pavements. Today, although measurements exist, it is essentially the reference r-tables that are easily accessible and therefore known by the actors

1 AITF (Association des Ingénieurs Territoriaux de France)

2 AFE (Association Française de l'Eclairage)

3 Office des Asphaltes, ROUTES DE FRANCE, SPECBEA (Spécialistes de la Chaussée en Béton Et des Aménagements), CIMbéton (Centre d'information sur le ciment et ses applications), EUROBITUME

4 Cerema (Centre d'Etudes et d'expertise sur les Risques, l'Environnement, la Mobilité et l'Aménagement), CERIB (Centre d'Etudes et de Recherches de l'Industrie du Béton), CTMNC (Centre Technique des Matériaux Naturels de Construction).

concerned. Moreover, when they are carried out, photometric measurements are generally made very quickly after the pavement has been installed, whether or not it has received a surface finish. Finally, the quality of public lighting is rarely assessed by taking into account the actual photometric properties of the pavement (Chain 2007).

Basics knowledge

Road photometry

The surface of a pavement is classified according to its reflective properties (CIE 144, 2001). The most characteristic parameter is the luminance coefficient q , given as:

$$q(\alpha, \beta, \gamma) = L(\alpha, \beta, \gamma) / (E_h) \quad (1)$$

It is the ratio of the observed luminance L in cd/m^2 , which the observer sees, to the illuminance E_h in lux, which is incident on the surface. The standard observation height is 1.5 m and the observation angle α is constant at 1° , which corresponds to an observation distance of 86 m. The lighting standards use the area of the road between 60 m and 160 m in front of the driver, as it is considered an important area obstacle detection. It has also been defined for interurban driving where the speed is about 80 to 90 km/h.

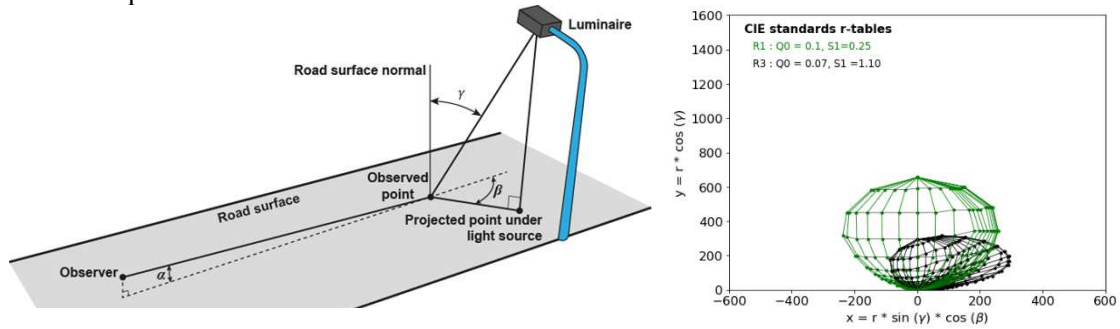


Figure 1 – Left: Road photometry characteristics angles, observation (α), deviation (β) and incidence (γ) Right: Representation of a the pavement reflection indicatrix for the standards r-tables R1 and R3

Since the 1980s, for practical reasons the luminance coefficient was replaced by the reduced luminance coefficient r in $\text{cd/m}^2/\text{lux}$, which is derived from q :

$$r(\beta, \gamma) = 10^4 \times q(\beta, \gamma) \times \cos^3 \gamma \quad (2)$$

A reduced coefficient table called r-table was defined, where the luminance coefficient r is given for a combination of fixed illumination angles β and γ . This table can be represented by a reflection indicatrix as shown in Figure 1. To simplify the description of photometric performance of road surfaces, the additional parameters Q_0 and S_1 are calculated. The average luminance coefficient Q_0 , represents the degree of lightness of the measured surface. It is calculated as the average of the luminance coefficients over the specified solid angle, Ω_0 . The specular factor S_1 represents the degree of specularly (shininess) of the observed surface. It is defined as the ratio of the reduced luminance coefficients of two specific illumination conditions.

$$Q_0 = \frac{1}{\Omega_0} \int q \, d\Omega \quad S_1 = \frac{r(\beta=0, \tan \varepsilon=2)}{r(\beta=0, \tan \varepsilon=0)} \quad (3) \text{ \& \; } (4)$$

The CIE (International Commission on Illumination) has defined different set of standard r-tables that are directly available in all lighting design software (CIE 066, 1984). Since 2001, the CIE (CIE 144, 2001) recommends a scaling of the chosen standard table according to the average luminance coefficient parameters Q_0 . It is obvious that the design of a road lighting system should be based on the knowledge of the actual luminance coefficient for the actual road. Because the actual quantity of q is not known, nor is listed as reference values in the EN standard (it provides only the directions in which q should be known), designers use in the calculations as q reference values the ones given in CIE 144 scientific publication. In France the CIE r-table type R3 is mostly used. In some countries (USA, UK, etc.), R1 reference table is used for cement concrete.

Road lighting

Guidelines and road lighting standards in Europe give values for illuminance and luminance and their distribution on the road surface according to a grid of points whose number N depends on the pole spacing S and number of traffic lanes (Figure 2).

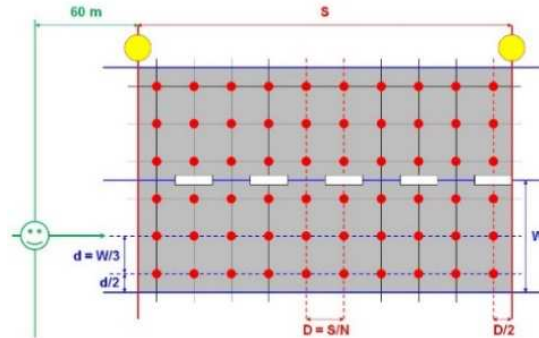


Figure 2 - Example of a calculation /measurement grid points (in red) in road lighting design/evaluation, according to the standard (w is the traffic lane width, d the transversal distance between two lines of points and D the longitudinal distance between two columns of points).

The standard road lighting quality criteria (CIE 140, 2000, CIE ILV, 2015) are derived from the luminance calculated on the grid, which are:

- The average luminance L_{ave} , calculated on every point of the grid
- Overall uniformity of luminance U_0 , as the ratio between the minimum and the mean
- Lengthwise uniformity of luminance U_l , as the ratio between the minimum and the maximum along the axis of each driving lane;
- Threshold Increment TI, as the measure of the disability glare defined in (Anon., 2011).
- Edge Illuminance Ratio EIR, as the ratio of the illuminance on a strip adjacent to the road, to the illuminance on the same strip lying on the adjacent driving lane.

The luminance is calculated from the axis of each lane of the road, embracing the whole width of the grid. For luminance, uniformities and edge illuminance ratio, the lowest value from all lanes is operative. For threshold increment, the highest value is operative.

This approach makes it possible to choose the lighting fixtures used and thus has an impact on urban energy consumption.

Methodology of the study

Pavements library constitution

A representative panel of 39 urban and interurban paving materials including innovative French technologies was first established and is described in Table 1.

Table 1 – Description of the 39 formulas studied

	Families	Aggregates size	Aggregates color	Binder	Surface finishes
Asphaltic concrete	4 medium coarse asphalt concrete ("BBSG") 8 very thin asphalt concrete ("BBM")	9 formulas in 0/10 3 formulas in 0/6	2 dark 10 light	8 bitumen 4 synthetic	9 raw surfaces 3 sandblasted
Mastic asphalt	2 sidewalk asphalt, AT 4 road asphalt, AC2Gr	2 formulas in 0/6 4 formulas in 0/10	6 light	3 bitumen 3 synthetic	2 raw surface, 2 shot blasted 2 sanded
Poured cement concrete	Not specified	6 formulas in 6/14 3 formulas in 4/16 1 formula in 0/10	7 light 3 dark	Cement	1 raw, 1 swiped, 1 smoothed, 2 deactivated, 2 bush hammered, 1 sanded, 2 sandblasted
Precast cement concrete	Not specified	5 formulas in 0/2 1 formula in 0/3 1 formula in 0/14	1 white 5 grey 1 black	Cement	2 raw, 1 washed, 1 shot blasted, 1 aged, 2 flamed
Natural stones	3 granite 1 limestone	Not applicable	Not applicable	Not applicable	4 flamed

Gonioreflectometer test campaign

Device presentation

The Cerema gonioreflectometer in Clermont-Ferrand measures the reflection of road surfaces under the observation conditions of a motorist at different observation angles including 1° , according to CIE specifications. It gives the 580 reflected luminance coefficients $q(\beta, \gamma)$ of the table, and the parameters Q_0 and S_1 . The mechanical positioning unit consists of a steel base on which is adapted a rotating measuring arm to change the sight angle β from 0 to 180° and a light source positioning system to vary the angle of incidence of light γ from 0 to 90° (see picture of Figure 3a). The mechanical movement system of the lamp and the arm carrying the photometer is computer controlled, so the data acquisition is fully automated.

The reference source is a type A halogen lamp (Philips PAR38 Spot bulb with a power of 120W, a nominal voltage of 24 V, a nominal flow rate is 1545 lm) with a color temperature of 2700K (warm light) and an angle of diffusion of 10° . The light source is fixed on a metal arc with a radius of 2.05 m, whose movement corresponding to the incidence angle is ensured by a motor connected to an indexer-transformer.

The illuminated area is a 10×10 cm square, always at the center of the sample. The “sample holder device” consists of a turntable allowing adjusting the height of the sample, its lateral positioning and inclination to obtain the horizontality of the sample upper surface (Figure 3b). This allows the measurement area to be centered on the sample without having to change the luminance meter and source settings.

Test campaign presentation

Sample pairs corresponding to the 39 selected surfacings described above were made. All the samples were then measured with the Cerema laboratory gonioreflectometer (Figure 2) in order to evaluate their photometric properties: table-r, Q_0 , S_1 (Muzet, Colomb, 2019).

All the pavements were first measured at their initial state (called T0). Half of the samples were then stored in fridges and half of them were installed outside in order to undergo a natural ageing during 30 months (called T30, Figure 3). This ageing implied sun, rain and any other weather condition exposition but not any mechanical influence such as traffic. The ageing applied corresponds to what could be observed on both the central lanes of roads and on urban pavements not used by cars, such as cycle paths, sideways or squares. It should be noted that this methodology does not take into account the possible evolution of the surface condition due to the use of the coating: wear, dirt, cleaning, etc.



Figure 3 – Left: the Cerema gonioreflectometer. Right: the samples outdoor during natural ageing

Lighting simulation parameters

The next step of the study presented here consisted in the realization of simulations of public lighting using the various table-r measured previously. The quality of lighting was designed and/or analyzed according to the criteria defined in the standard (EN 13201, 2015).

In all the cases presented in this study, the simulations correspond to a frequent situation where the dimensioning of the lighting is done for an already existing infrastructure, and there is a change of the luminaires called relamping. Thus the geometry of the lighting and the road are imposed.

The first scenario is the most frequent case where the lighting is dimensioned without knowing the photometric characteristics of the pavement in place. The lighting design was conducted using an R3 table and an R1 table, giving two possible LED luminaire. To study the impact of the real pavement photometry, lighting performance simulation were conducted by considering the photometric characteristics measured on the samples at their initial state (T0) and after ageing (T30) with the use of the LED luminaire corresponding to R3 and R1 pavement. The compliance with the lighting standard requirement in term of luminance level and homogeneity was studied for the two LED luminaire chosen with the R3 and R1 pavement.

The second scenario of the simulation consisted in designing the lighting directly according to the real photometric characteristics measured on the samples, T30 measurements were used.

The two scenarios were applied to a residential street composed of two driving lanes, illuminated by a single-sided lighting installation.

Results

Photometry results

The graph of the Figure 4 presents the evolution of Q_0 and S_1 parameters during aging for each sample, the origin of the arrow is the T0 measurement and the arrowhead is the T30 measurement. Each sub-figure corresponds to a specific pavement type.

The results concerning natural stones are presented for information in the graph corresponding to the precast concrete samples but are not studied hereafter because only the results at T0 are available for the moment, the measurements of T30 will be made in summer 2022. Although 12 asphalt concrete samples were made, several were degraded and had to be remade and their results are therefore not presented here.

General analysis

The results of the photometric measurements of the pavements show that there is an enormous diversity of behaviors. This is especially true at T0 but remains true after 30 months of aging. These results confirm again that the 4 CIE standard tables do not reflect the diversity of all pavements (Dumont et al., 2007, Gidlund et al., 2019).

It can be observed that in each sub-figure, all arrows are pointing downwards, which means that there is a decrease in specularity over time on the vast majority of pavements. This evolution is really important for asphalt concrete and mastic asphalt families which were initially very specular. It remains for cement concrete samples, both poured and precast, but this phenomenon is less pronounced. These results are in accordance with previous results (Dumont et al., 2007, Muzet and Abdo, 2018, Muzet, Greffier, et al., 2019).

After 30 months, all samples are classified as CIE R1 type (except 2 which are R2 type), mainly due to this specularity evolution. Such low values of specularity are not usually measured on conventional circulated road surfaces (Dumont et al., 2007, Gidlund et al., 2019) but it was shown in (Jackett and Frith, 2010) that non circulated pavements were more diffuse after aging than the ones exposed to traffic.

The evolution with time of the average luminance coefficient Q_0 also differs depending on the pavement surface material. Except for initially very specular surfaces, for the asphaltic concrete and mastic asphalt samples, the luminance coefficient increased between T0 and T30. A general explanation could be that ageing with meteorological condition has a kind of tarnishing effect on these samples, slightly lightening their surface. On the contrary, the luminance coefficient of cement concrete almost systematically slightly decreased. Since the surface of these samples was quite light from the start, ageing seems to darken them slightly, which is rather

confirmed by the visual evolution. As above, this behavior is not observed when the T0 samples were more specular.

At T30, all the concrete and precast concrete samples have characteristics close to the R1 reference table, this is particularly marked for precast concrete.

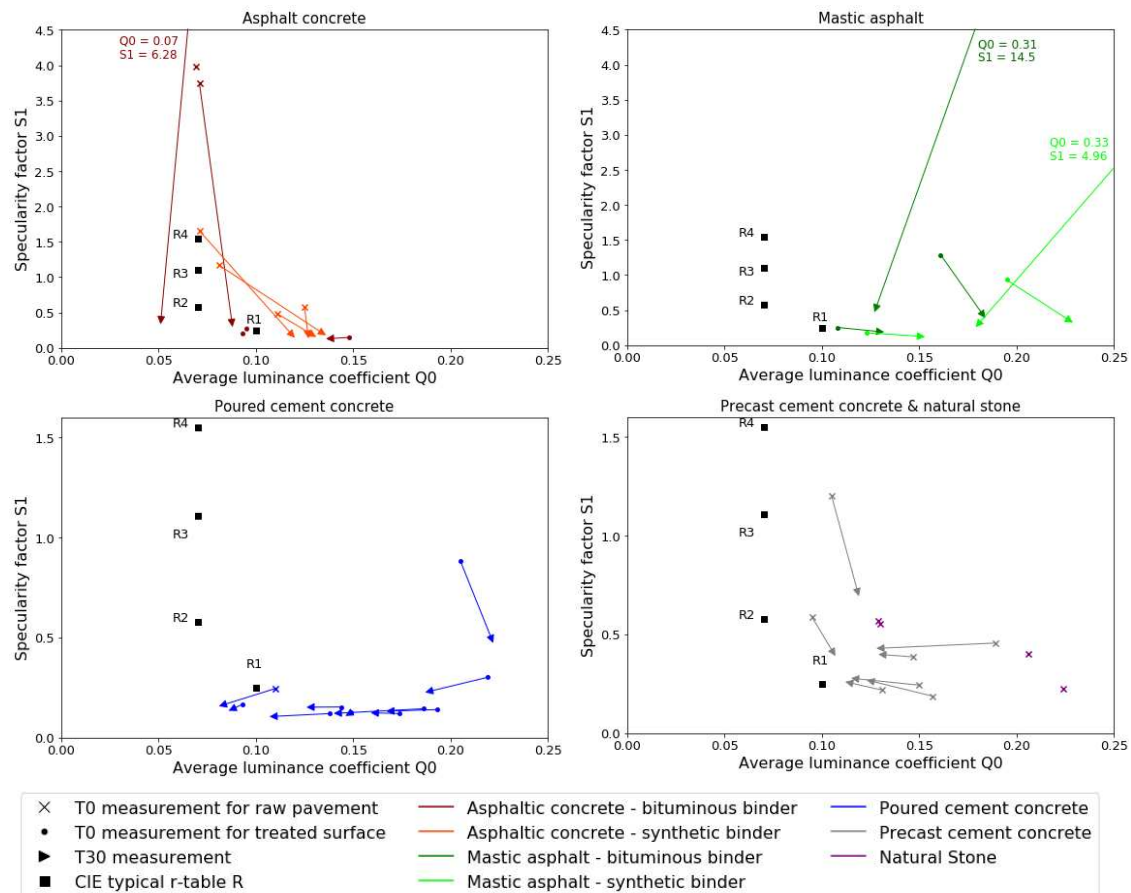


Figure 4 - Representation of the P&L photometric measurements (S_1 and Q_0) for all the pavements. T0 is the origin of the arrow and T30 measurement corresponds to the arrowheads. The reference CIE r-tables are also presented in black squares. The S_1 scale is different between the two top and bottom graphs

Binder and finishes impacts

Some of the above observations can be explained by the photometric characteristics of the binder used. Indeed, bitumen is initially a very shiny and specular material, which however evolves very quickly. This behavior remains true for synthetic binders although it is attenuated, in this case they are less specular in the initial state and evolve less, their characteristics at T30 is then comparable to that of bitumen.

Both for cast and precast concrete, samples made with cement show little change over time. This is due to the fact that the cement has a diffuse appearance almost from the beginning and does not change much.

The color of the binder can also explain the evolution of Q_0 , except in cases where the specularity was very high at the beginning. Indeed the bitumen is very dark while the cement is clear.

Samples with a specific surface finishes are marked on the graphs with a circle instead of a cross. It can be seen that for both asphalt concrete and mastic asphalt samples, the specularity values are systematically lower for the ones with a specific surface finishes (sanded, sandblasted or shot blasted). In accordance with what was said earlier, the photometric properties of these samples evolve less over time than the others.

There is no observable difference between the finishing states for the poured cement concrete samples.

Lighting simulation results

Tables presentation

The tables in Figure 5 show the results of all the simulations (scenarios 1 and 2). Each table corresponds to a family of pavements. Natural stones are not presented because all the measurements have not yet been performed. For the scenario 1, the red boxes correspond to cases where the designed lighting would not meet the public lighting quality standards. For the scenario 2, since the lighting design considers the measured pavement photometry, the standard lighting quality requirement are always respected, so the difference of the installed luminous power with a R3 pavement is given in percentage (and therefore the potential economy in power). As previously, only 7 of the 12 asphalt concrete samples are presented here.

						Scenario 1		Scenario 2
	Sample Nb.	Agg. Color.	Surf. Fin.	Bind.	Age	R3	R1	Power evolution
Asphalt concrete	1	D	-	B	10 1.30			69%
	5	L	-	B	10 1.30			-13%
	6	L	SB	B	10 1.30			-53%
	9	L	-	S	10 1.30			-46%
	10	L	-	S	10 1.30			-40%
	11	L	-	S	10 1.30			-51%
	12	L	-	S	10 1.30			-44%
						Scenario 1		Scenario 2
	Sample Nb.	Agg. Color.	Surf. Fin.	Bind.	Age	R3	R1	Power evolution
Mastic asphalt	1	L	-	B	10 1.30			-30%
	2	L	ShB	B	10 1.30			-61%
	3	L	Sd	B	10 1.30			-45%
	4	L	-	S	10 1.30			-58%
	5	L	ShB	S	10 1.30			-57%
	6	L	Sd	S	10 1.30			-56%
						Scenario 1		Scenario 2
	Sample Nb.	Agg. Color.	Surf. Fin.	Bind.	Age	R3	R1	Power evolution
Poured cement concrete	1	L	Sw	C	10 1.30			-51%
	2	L	Sm	C	10 1.30			-60%
	3	D	Dea	C	10 1.30			-19%
	4	L	Dea	C	10 1.30			-51%
	5	D	BH	C	10 1.30			-43%
	6	L	BH	C	10 1.30			-62%
	7	L	Sd	C	10 1.30			-54%
	8	D	SB	C	10 1.30			-36%
	9	L	SB	C	10 1.30			-47%
	10	-	-	C	10 1.30			-26%
						Scenario 1		Scenario 2
	Sample Nb.	Agg. Color.	Surf. Fin.	Bind.	Age	R3	R1	Power evolution
Precast cement	1	-	-	C	10 1.30			-46%
	2	-	W	C	10 1.30			-25%
	3	-	ShB	C	10 1.30			-34%
	4	-	Ag	C	10 1.30			-20%
	5	-	-	C	10 1.30			-33%
	6	-	Fl	C	10 1.30			-33%
	7	-	Fl	C	10 1.30			-33%

Abbreviations :

- D: dark
- L: light
- B: bitumen
- S: synthetic
- SB: sand blasted
- ShB: shot blasted
- Sd: sanded
- Sw: swiped
- Sm: smoothed
- Dea: deactivated
- BH: bush hammered
- W: washed
- Ag: aged
- Fl: flamed

Figure 5 – Simulation results for each pavements families. For the scenario 1, a check of compliance with the standard requirement is conducted using R3 and R1 typical r-table (green color). For the scenario 2 that uses the T30 measured photometry, the difference of power consumption is computed in %.

Scenario 1 observations

In the case of scenario 1, i.e. following the usual methodology of designing the lighting on the basis of the references table-r, the simulations show that the quality of the lighting actually obtained with the measured photometric properties would not comply with the standards for a large quantity of pavements. This is especially true when the lighting designing is carried out with the R3 r-table where many boxes appear in red.

The results for asphalt concrete show that the use of a synthetic binder (samples 9 to 12) leads to a more standard-compliant lighting. This is rather logical since their photometric characteristics are closer to the reference table-r (Figure 4). The results for mastic asphalt show a strong impact of surface finishing (shot blasting and sand blasting, samples 2, 3, 5 and 6) resulting

in greater compliance with the standards. The results for cast concrete samples systematically show a non-conforming lighting compared to the standards. Although not presented here, the values show in fact that non-compliance with the standard is almost systematically due to poor lengthwise uniformity of luminance, a direct consequence of over-illumination (average luminance too high). This is consistent with the photometric properties of the latter, namely a high Q_0 above R1 logically inducing high luminance values.

The results for the precast concrete samples show compliance with the standard in a large majority of cases. This can be explained by the proximity of the photometric properties of these pavements to the reference r-tables, especially R1 and after samples aging, as shown in Figure 4.

Scenario 2 observations

In the case of scenario 2, i.e. the design of the lighting based on the actual photometric properties measured on the samples, the standards are by definition systematically met. The results show very significant decreases in power, often between 30 and 60%. It is interesting to note that this gain remains valid even in cases where the standard was already met with scenario 1 (e.g. for precast concrete).

Conclusions

This study allowed the realization of photometric measurements on a large number of classical and innovative French urban surfaces. In particular, this allowed us to obtain results on pavement families not well represented in the literature, such as mastic asphalt, precast concrete or natural stone. Beyond the raw results, it was also an opportunity to highlight the impact of aging due to weather. The results show a very large variability and largely confirm the lack of relevance and representativeness of the CIE reference r-tables.

The simulations showed first of all that the classical method of lighting design based on the CIE references r-tables is limited. Indeed, it leads in many cases to a non-conformity with the standard. This is particularly pronounced for poured concrete, asphalt concrete and mastic asphalt samples. Nevertheless, the use of synthetic binder instead of bitumen had a positive impact.

The use of a surface treatment also has a beneficial effect on the asphalt concrete and mastic asphalt samples. No effect could be observed for poured cement concrete. This seems logical given the photometric properties of bitumen and its impact on the specularity of samples at T0.

Finally, the simulations performed by dimensioning the lighting according to the real photometric properties of the surfacings show a strong reduction of the luminous power to be installed in order to respect the standard. In almost all cases, this reduction is of the order of 30% to 60%, including when the standard was already met using the reference r-table. Similar simulations have been performed on 4-lane road profiles under interurban conditions, and the results also show the same trends (Muzet V. 2021).

This shows the importance of using the real pavement properties and thus making measurements. This ensures that the standard is met and thus avoids comfort and safety problems, but also results in significant energy savings.

The results also showed the importance of not performing these measurements of the properties of the samples in their initial state but rather to wait for a certain ageing. On this point, the bibliography shows a rather fast evolution in the first months after the implementation. In order to mitigate this phenomenon, the results also show the beneficial impact of surface treatments for bitumen-based formulas. These treatments at least prevented too great an evolution of the photometric properties over time but also allowed a better respect of the standard.

The library of pavements developed within the framework of the Pavements & Light working group shows its full interest here. It offers an overview of the photometric properties of a large number of different pavements and will be a source of information on them. It also demonstrates the interest in changing urban lighting design practices for multiple interests: user comfort and safety, but also optimization of energy consumption by managers.

Although not presented here, the work carried out on the different pavement samples also provides elements on other observation angles that could be relevant in an urban environment. Similarly, colorimetric and radiometric measurements have been carried out that could provide elements of answers for treating urban heat island phenomena (Lebouc L. 2022).

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